

Feedback Control Analysis of a Buck Converter

1. Objective:

The goal of this project is to design and evaluate a closed-loop voltage controller for a buck converter. (1) Derive the small-signal control-to-output transfer functions of DC-DC converters. (2) Design a feedback controller to regulate the output voltage. (3) Use PLECS simulations to validate the controller. (4) Investigate how controller parameters affect stability, transient response, and steady-state error.

2. System Description

Parameter	Value
Input Voltage V_{in}	24 V
Output Voltage V_{out}	12 V
Switching Frequency	200 kHz
Inductor L	50 μ H
Capacitor C	100 μ F
Load Resistance R	10 Ω
Steady-State Duty Ratio D	0.5

Table 1. Converter Design parameters.

Component Selection: *MOSFET*: Infineon IRLML6344TRPBF (N-ch, 5 A, 30 V, SOT-23. $R_{on} \approx 37$ m Ω , low gate charge, fast switching — suitable for 200 kHz). *Diode*: Schottky SS32–S310 ($V_f \approx 0.85$ V, fast recovery, low reverse-recovery loss ideal for asynchronous rectification at high frequency).

3. DC-DC Converter Modeling and Transfer Functions

3.1. State-Space Model -Buck:

$$v_L = L \frac{di_L}{dt}; \quad i_c = C \frac{dv_c}{dt}$$

$$\text{Switch On Diode off } 0 < t < DT: -v_L = v_c; \quad i_{in} = i_L = i_c + i_o$$

$$\text{Switch Off Diode on } DT < t < T: v_L = -v_c; \quad i_L = i_c + i_o$$

$$\text{Averaged Model: } [x' = Ax + Bu]; [y = Cx + Du]$$

$$\frac{di_L}{dt} = \frac{1}{L} (v_{in} - v_c)d - \frac{v_c}{L} (1 - d) = -\frac{v_c}{L} + \frac{dv_{in}}{L}$$

$$\frac{dv_c}{dt} = \frac{1}{C} (i_L - i_o)d + \frac{1}{C} (i_L - i_o)(1 - d) = \frac{i_L}{C} - \frac{i_o}{C} = \frac{i_L}{C} - \frac{v_c}{RC}$$

$$x' = \begin{bmatrix} i_L' \\ v_c' \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_c \end{bmatrix} + \begin{bmatrix} \frac{d}{L} \\ 0 \end{bmatrix} [v_{in}]; \quad y = v_o = v_c = [0 \quad 1] \begin{bmatrix} i_L \\ v_c \end{bmatrix}$$

3.2. Small Signal Linearization

$$v_{in} = V_{IN} + \hat{v}_{in}, \quad d = D + \hat{d}, \quad v_c = v_o = V_O + \hat{v}_o$$

$$dv_{in} = (D + \hat{d})(V_{IN} + \hat{v}_{in}) = DV_{IN} + D\hat{v}_{in} + \hat{d}V_{IN} \text{ (2nd order terms } \approx 0)$$

$$\text{linearized inductor equation: } i_L' = \frac{-\hat{v}_o}{L} + \frac{D\hat{v}_{in}}{L} + \frac{V_{IN}\hat{d}}{L}$$

$$\text{Applying the Laplace transform to the averaged model } X = (sI - A)^{-1}BU, Y = C(sI - A)^{-1}BU + DU$$

3.3. Control-to-Output Transfer function:

For controller design, the input voltage is assumed constant, $\hat{v}_{in} = 0$.

Feedback Control Analysis of a Buck Converter

$$G_{vd}(s) = \frac{Y}{U} = \frac{M\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = [0 \quad 1] \begin{bmatrix} \frac{s + \frac{1}{RC}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} & \frac{\frac{1}{L}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \\ \frac{\frac{1}{C}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} & \frac{s}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \end{bmatrix} \begin{bmatrix} \frac{V_{in}}{L} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{V_{in}}{LC} \\ \frac{1}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \end{bmatrix} = \left[\frac{4.8 \times 10^9}{s^2 + 10^3 s + 2 \times 10^8} \right]$$

This transfer function represents the small-signal relationship between duty-cycle perturbation and output voltage and is therefore used for controller design.

3.4. Line-to-Output Transfer function:

To study input-voltage disturbances, the duty cycle is held constant, $\hat{d} = 0$.

$$G_{vg}(s) = \frac{Y}{U} = \frac{M\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = [0 \quad 1] \begin{bmatrix} \frac{s + \frac{1}{RC}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} & \frac{\frac{1}{L}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \\ \frac{\frac{1}{C}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} & \frac{s}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \end{bmatrix} \begin{bmatrix} \frac{d}{L} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{d}{LC} \\ \frac{1}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \end{bmatrix} = \left[\frac{10^8}{s^2 + 10^3 s + 2 \times 10^8} \right]$$

This transfer function describes how disturbances in the input voltage propagate to the output when the duty cycle is held constant. It is useful for analyzing line regulation but is not the transfer function used to design the feedback controller.

Parameter	Formula	Value	Physical Meaning
Natural frequency ω_n	$\frac{1}{\sqrt{LC}}$	14142 rad/s	Resonant frequency of the LC filter
Damping ratio ζ	$\frac{1}{2RC\omega_n}$	0.0354	Measure of how quickly oscillations decay
Control-to-output DC gain M_{vd}	V_{in}	24	Duty-cycle-to-output sensitivity
Line-to-output DC gain M_{vg}	$D = \frac{V_{out}}{V_{in}}$	$\frac{12}{24} = 0.5$	Vin-to-output sensitivity
System order	—	2nd order (L & C)	Independent energy-storage elements
Poles	$s^2 + \frac{s}{RC} + \frac{1}{LC} = 0$	$-500 \pm 14142i$	Determines causing the system's response to grow, decay, or oscillate

Table 2. Small signal parameters.

If R grows, $R \rightarrow \infty$; $\zeta = \frac{1}{\omega_n \times 2RC} = 0$; $\omega_n = \sqrt{\frac{1}{LC}}$; **damping ratio approaches zero, natural frequency remains the same unaffected by R**

4. Open Loop PLECS Simulation of the System

Feedback Control Analysis of a Buck Converter

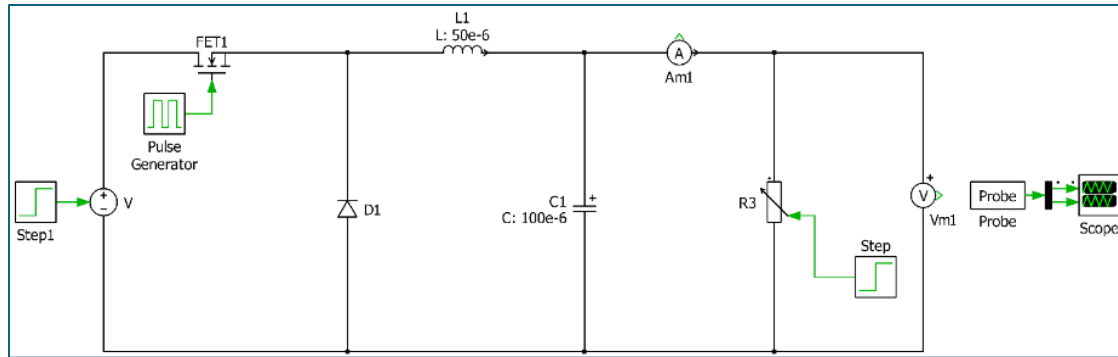


Figure 1. PLECS open loop schematic for buck converter

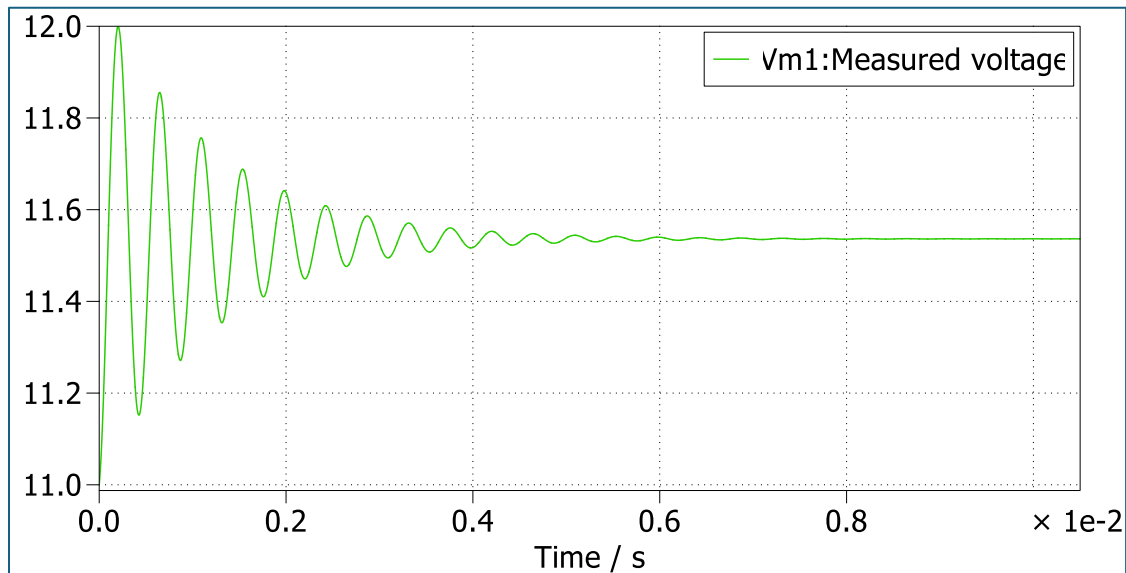


Figure 2. Output voltage of open loop buck converter

Simulated Output voltage is $\approx 11.54\text{V}$ vs ideal 12V caused by MOSFET on-resistance drop ($\approx 37\text{ m}\Omega \times 1.2\text{ A} \approx 41\text{ mV}$) and diode forward voltage ($V_f \approx 0.85\text{ V}$ at 1.2 A) and the settling time is 6ms , maximum overshoot is 12.00V and undershoot is 11.15V , deviation = 0.46 V

Transient Tests

- Apply a step change in load at $t=0.01\text{s}$: $10\ \Omega \rightarrow 5\ \Omega$

Feedback Control Analysis of a Buck Converter

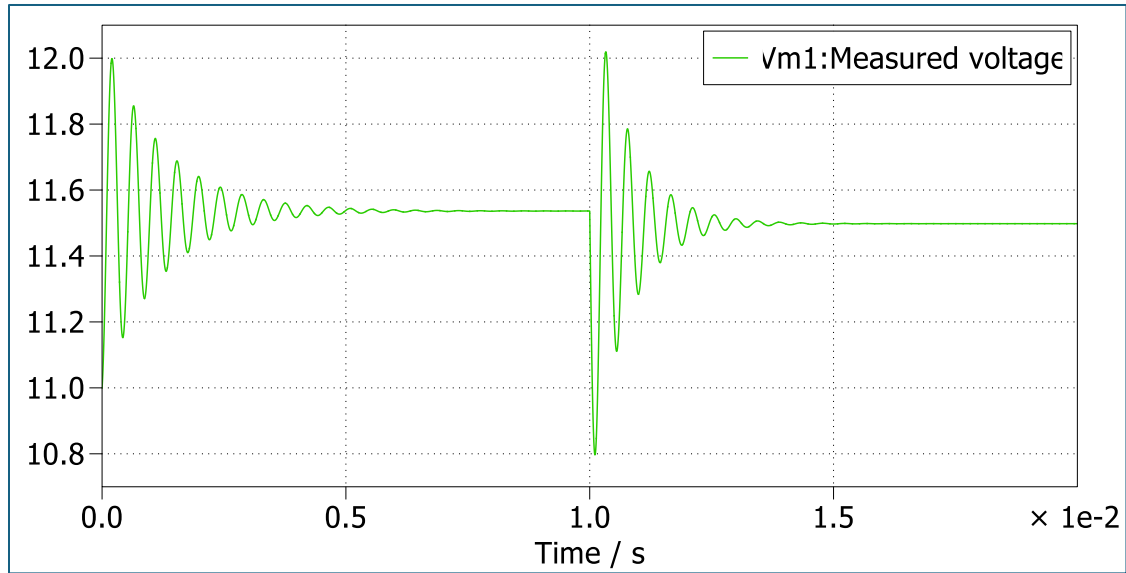


Figure 3. Step change in load $10\ \Omega \rightarrow 5\ \Omega$ at $t=0.01\text{s}$

ii. Keep the resistor at its original value, change the voltage from 24V to 20V as a step change:

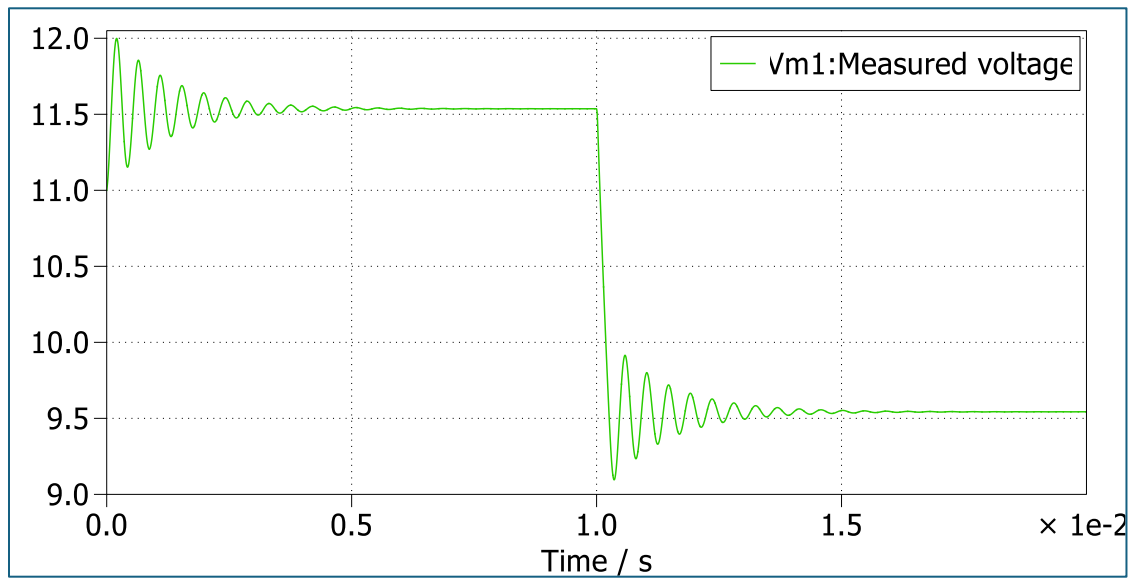


Figure 4. Step change in voltage $24\text{ V} \rightarrow 20\text{ V}$ at $t=0.01\text{s}$

Test	V_{out} before step average	V_{out} after step average	Max OS	Max US	Transient dip	Deviation (from 12V)	Settling Time	% OS
Load step: $10\ \Omega \rightarrow 5\ \Omega$	11.54V	11.50 V	12.00V	10.73V	-0.81	-0.50 V	5 ms	4.6%
Vin step: $24\text{ V} \rightarrow 20\text{V}$	11.54V	9.54 V	9.9 V	9.10 V	-0.44	-2.46 V	6 ms	3.9%

Table 3. Open loop parameters.

Feedback Control Analysis of a Buck Converter

Q1 Voltage deviation: Load step: -0.81V transient dip, settling at -0.50V from 12V, due to increased $\frac{R_{on}}{V_f}$ drops at higher load current. V_{in} step: permanent drop to 9.54V — well beyond the loss-only degradation, since open-loop control cannot track a changed V_{in} at all.

Q2 Why open-loop fails: No error sensing — D is a fixed parameter, not a function of V_{out} . Any disturbance in V_{in} or load directly propagates to V_{out} with no corrective mechanism. Sensitivity to operating-point variation — The sensitivity function $S(s) = \frac{1}{1 + G_c G_{vd}} = 1$ for open-loop ($G_c = 0$). This means output disturbances are transmitted with unity gain — there is zero rejection of load/supply perturbations. The ideal $V_{out} = DV_{in}$ ignores switching, conduction, ESR, inductor, and diode losses. Since $D = 0.5$ is calculated without accounting for these losses, steady-state error is structurally unavoidable in open loop. Output impedance: The open-loop output impedance $Z_o(s) = \frac{G_{vd}(s)}{G_{id}(s)}$ is set entirely by the LC filter.

5. Feedback Control

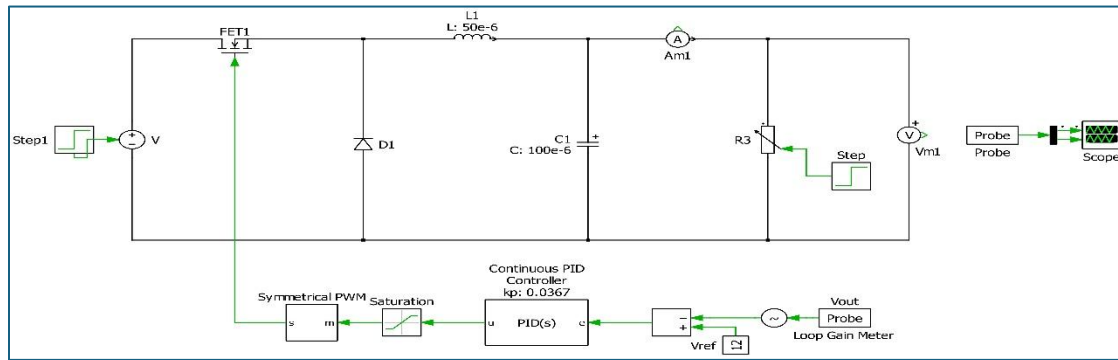


Figure 5. Closed loop buck converter circuit

Design rule used throughout: Crossover frequency $\omega_c = 0.1\omega_n = 1.4142 \times 10^3$ ensures bandwidth is well below LC resonance and switching frequency two decades below the switching frequency, ensuring adequate phase margin and preventing switching noise from entering the control loop.

5.1. Proportional Control

Control law: $d(t) = K_p(V_{ref} - V_o) = K_p(12 - V_o)$

Analytical design: enforce $|G_c(j\omega_c)G_{vd}(j\omega_c)| = |K_p G_{vd}(j\omega_c)| = 1$

At $\omega_c = 1414.2 \frac{rad}{s}$ $|G_{vd}(j\omega_c)| \approx 24$ (since $\omega_c \ll \omega_n$, plant magnitude $\approx$ DC gain).

Therefore $K_p = \frac{1}{G_{vd}(j\omega_c) \approx 24} = \frac{1}{24} = 0.0417$ (initial controller parameters obtained from the small-signal model).

Closed loop TF: $T(s) = \frac{G_c(s)G_{vd}(s)}{1 + G_c(s)G_{vd}(s)} = \frac{10^8 K_p}{s^2 + 10^3 s + 2 \times 10^8 + 10^8 K_p} = \frac{M \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$

New Closed loop parameters with $K_p = 0.042$: $\omega_{n,cl} = 20000$ rad/s; $\zeta_{cl} = 0.025$; $M = 24$

Task 1: PLECS implementation: Proportional gain block (K_p) receives error signal $e = V_{ref} - V_{out}$ from summing junction. Output passes through saturation block [0,1] to PWM carrier (200 kHz symmetrical). V_{out} measured via voltmeter probe feeding back to subtractor.

Task 2: Simulation:

Feedback Control Analysis of a Buck Converter

- i. Keep the resistor at its original value, change the voltage from 24V to 20V as a step change to evaluate controller performance, when a line disturbance is introduced.

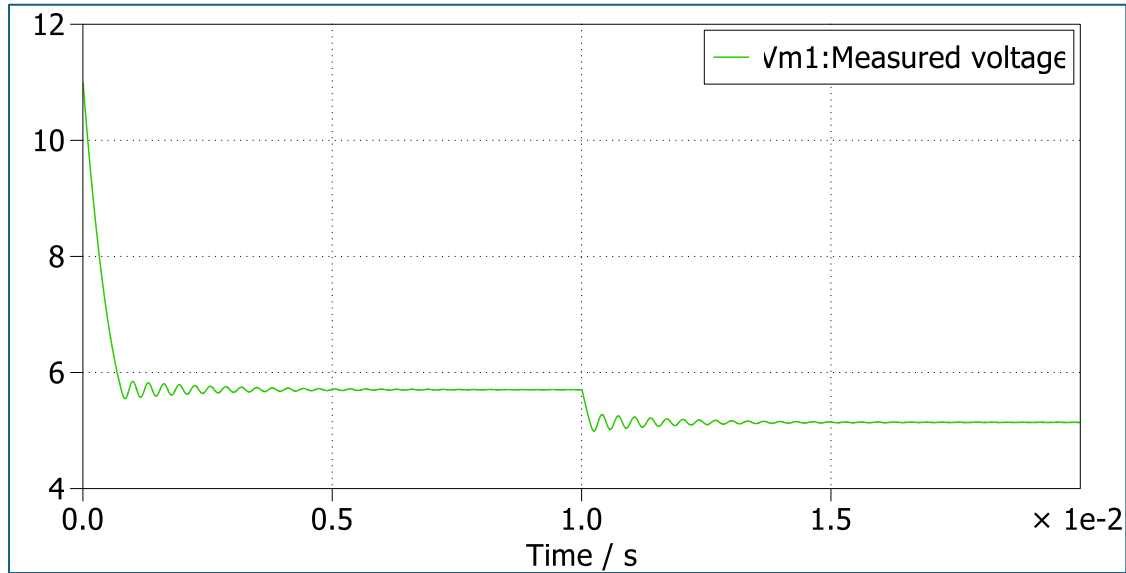


Figure 6. $K_p=0.042$ step change in voltage 24V to 20V at 0.01s

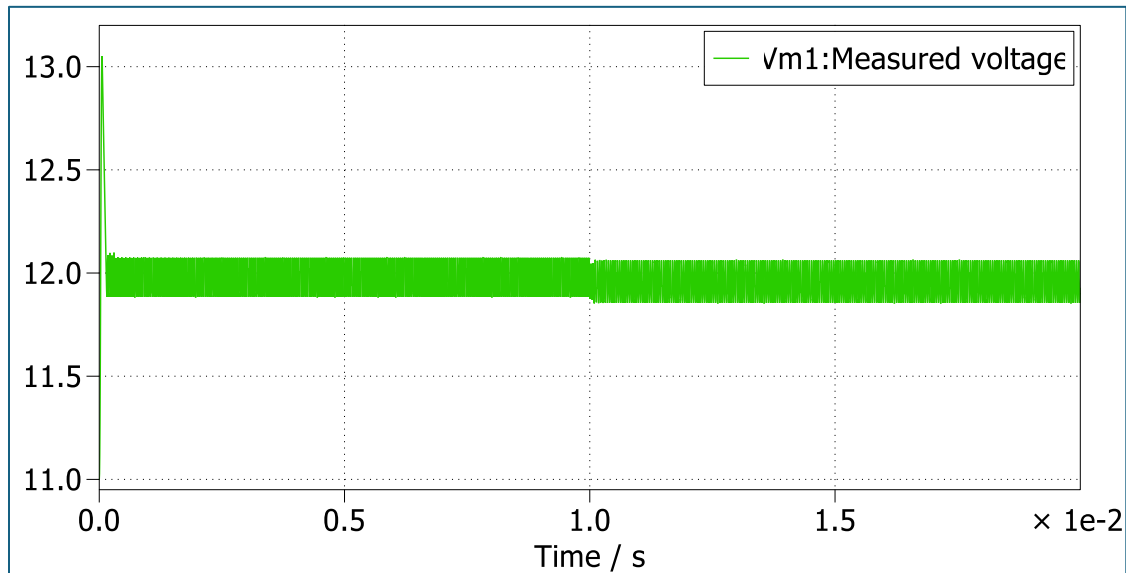


Figure 7. $K_p=11$, step change in voltage 24V to 20V at 0.01s

K_p	$V_{out,avg}$ before step	$V_{out,avg}$ after step	$V_{out,ripple}$ before	$V_{out,ripple}$ after	% OS	% US	Settling time	Comment
0.042	5.68 V	5.12 V	5.680- 5.679 = 0.001 V	11.60- 11.26 = 0.34 V	2.5%	2.9%	7.5ms	Insufficient control effort resulting in a large steady- state error and poor line regulation.
11	11.99 V	11.97 V	12.08- 11.89 = 0.18 V	12.07- 11.86 = 0.20 V	0%	0%	0s	Smaller SS error (0.04V); good tradeoff between speed and accuracy

Feedback Control Analysis of a Buck Converter

Table 4. *Vin-Step Results (24V→20V)*

- ii. Keep the voltage at its original value, change the resistance from $10\ \Omega$ to $5\ \Omega$ as a step change to evaluate controller performance, when a line disturbance is introduced.

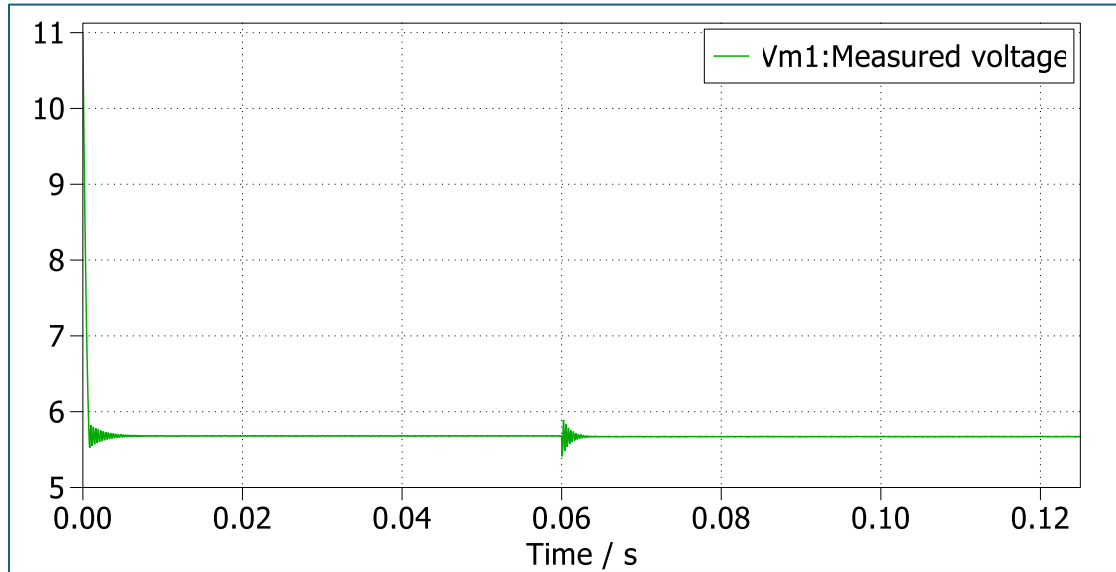


Figure 8. $K_p=0.042$ step change in load $10\ \Omega$ to $5\ \Omega$ at $0.06s$

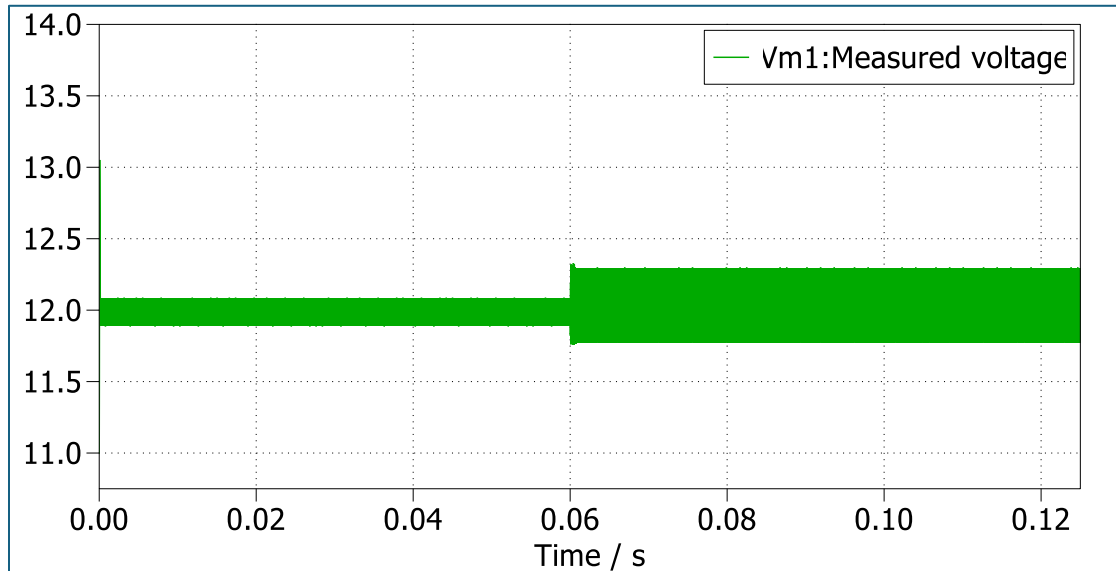


Figure 9. $K_p=11$ step change in load $10\ \Omega$ to $5\ \Omega$ at $0.06s$

K_p	$V_{out,avg}$ before step	$V_{out,avg}$ after step	$V_{out,ripple}$ before	$V_{out,ripple}$ after	% OS	% US	Settling time	Comment
0.042	5.68 V	5.67 V	5.680- 5.679 = 0.001 V	5.672- 5.671 = 0.001 V	0.04%	2.9%	7.0 ms	Fast but large SS error (6.32V from 12V); K_p too small

Feedback Control Analysis of a Buck Converter

11	11.99 V	11.97 V	12.08- 11.89 = 0.18 V	12.28- 11.78 = 0.50 V	0%	0%	0s	Smaller SS error (0.04V); good tradeoff between speed and accuracy
----	---------	---------	-----------------------------	-----------------------------	----	----	----	--

Table 5. Step load results $10\ \Omega \rightarrow 5\ \Omega$

Task 3: $s^2 + 10^3s + 2 \times 10^8 + 10^8K_p = 0$

K_p	Poles	$\omega_{n,cl}$	ζ_{cl}	T(0)	SS error
0.042	$-500 \pm j19994$	20000 rad/s	0.025	0.5	6.0 V
11	$-500 \pm j230217$	230217 rad/s	0.0139	0.996	0.046 V

Table 6. Eigen values of system at different K_p

As K_p increases, $\omega_{n,cl}$ rises and the poles move further along the imaginary axis, increasing oscillation frequency and reducing SS error. The real part $\sigma = -500$ stays fixed $\sigma = -\frac{1}{2RC}$, confirming the dominant decay rate is set by the power stage, not K_p .

Q1: Does output settle exactly at 12.00? No. $T(0) = \frac{K_p}{0.0417 + K_p} < 1$ always for P-control, Therefore, proportional control alone cannot guarantee perfect tracking of the reference voltage. Increasing (K_p) reduces the error but does not force it to zero.

Q2: Why steady-state error? A proportional controller produces $d = K_p \times e$. At steady state $d = D_0$, so $e = \frac{D_0}{K_p} \neq 0$. $T(0) < 1$ always — the controller gain is finite, so the output can never fully reach the reference without a residual error to drive it. Only an integrator can force $e \rightarrow 0$ at DC.

Settling time: The linearized model predicts that the dominant exponential decay is determined primarily by the converter dynamics. However, the nonlinear switching simulations show improved transient performance as K_p increases

5.2. PI Control

Control law: $d(t) = K_p e(t) + K_i \int e(t) = K_p(V_{ref} - V_o) + K_i \int (V_{ref} - V_o)$;

$$G_c(s) = K_p + \frac{K_i}{s} = K_p \frac{s+z}{s}; z = \frac{K_i}{K_p}$$

Step 1: $\omega_c = 0.1 \times \omega_n = \frac{1,414 \text{ rad}}{s}$

Step 2: $z = 0.3 \times \omega_c = 0.3 \times 1414 = 424 \text{ rad/s}$ (between $0.1-0.5 \times \omega_c$ — midpoint chosen)

Step 3: $|G_c(j\omega_c)| = K_p \times \frac{\sqrt{\omega_c^2 + z^2}}{\omega_c} = K_p \times \frac{\sqrt{1414^2 + 424^2}}{1414} = 1.044 K_p$

Step 4: $|G_c(j\omega_c) \times G_{vd}(j\omega_c)| = 1 \rightarrow 1.044 K_p \times 24 = 1 \rightarrow K_p = \frac{1}{24 \times 1.044} = 0.0399 \approx 0.04$

Step 5: $K_i = K_p \times z = 0.0399 \times 424 = 16.922 \approx 16.9$

$$T(s) = \frac{G_c(s)G_{vd}(s)}{1 + G_c(s)G_{vd}(s)} = \frac{10^8 K_p}{s^2 + 10^3 s + 2 \times 10^8 + 10^8 K_p + \frac{10^8 K_i}{s}} = \frac{10^8 s K_p + 10^8 K_i}{s^3 + 10^3 s^2 + s(2 \times 10^8 + 10^8 K_p) + 10^8 K_i}$$

$$s^3 + 10^3 s^2 + s(2 \times 10^8 + 10^8 K_p) + 10^8 K_i = 0; \quad s \approx -207.6, -396.2 \pm j19780$$

Simulation: Keep K_i fixed and tune K_p to investigate the effect of proportional gain

Feedback Control Analysis of a Buck Converter

i. Apply a step change in load at $t=0.01s$: $10\ \Omega \rightarrow 5\ \Omega$

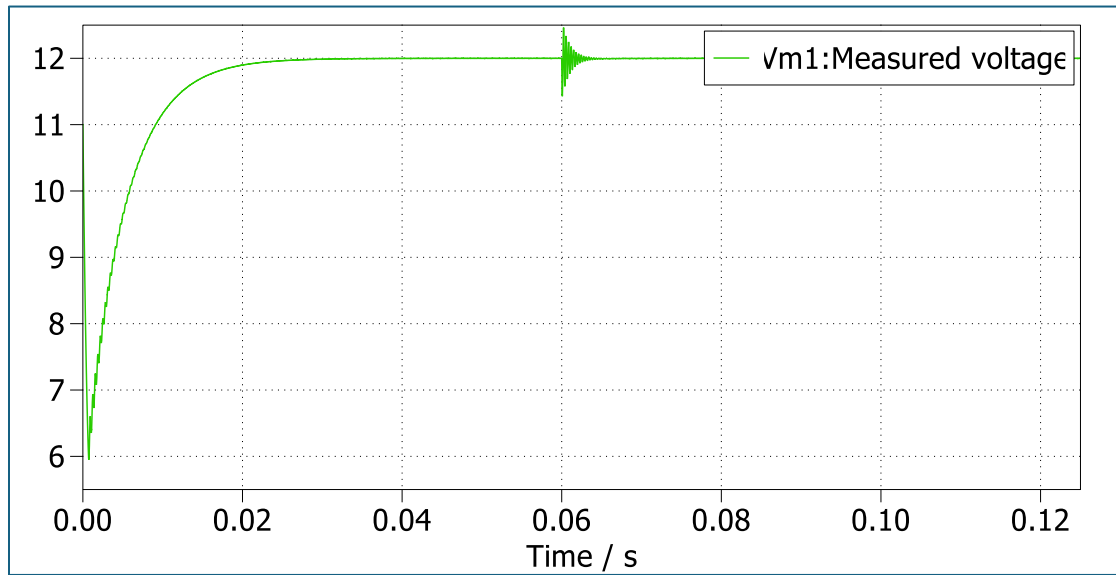


Figure 10. $K_p=0.04$, $K_i=16.9$, step change in load at $t=0.06s$

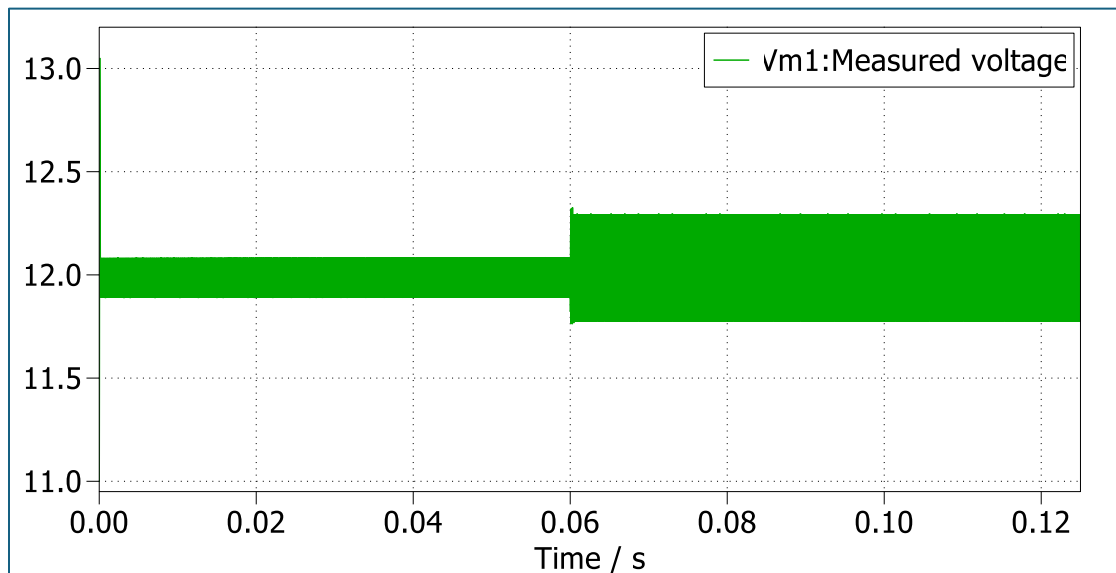


Figure 11. $K_p=11$, $K_i=16.9$, step change in load at $t=0.06s$

K_p	V_{out} before step (avg)	V_{out} after step (new SS avg)	V_{out} ripple before	V_{out} ripple after	% OS	% US	Settling time	Comment
0.04	12.00 V	12.00 V	12.15-11.84 = 0.31 V	12.17-11.84 = 0.33 V	0%	0%	6.7 ms	$z=424$, near ω_c — some aggression
11	12.00 V	11.96 V	12.08 - 11.90 = 0.18 V	12.35 - 11.76 = 0.69 V	2.5%	2.0%	1.73 ms	$z=77 \ll \omega_c$ — smooth regulation

Table 7. Comparison of bad and good K_p for load step

Feedback Control Analysis of a Buck Converter

ii. Keep the resistor at its original value, change the voltage from 24V to 20V as a step change:

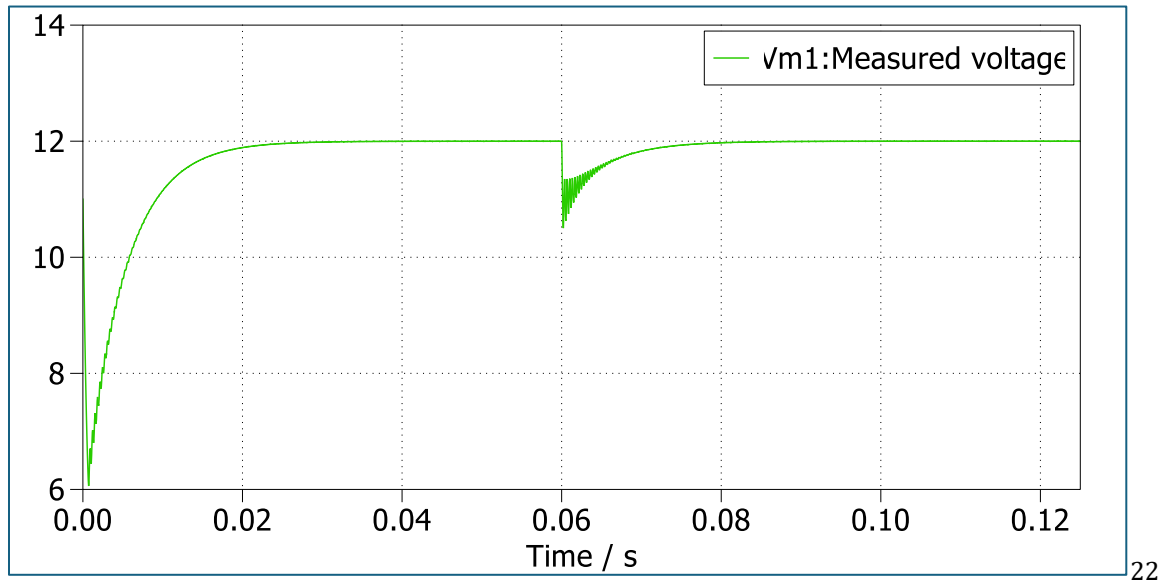


Figure 12. $K_p=0.04$, $K_i=16.9$, step change in voltage at $t=0.06s$

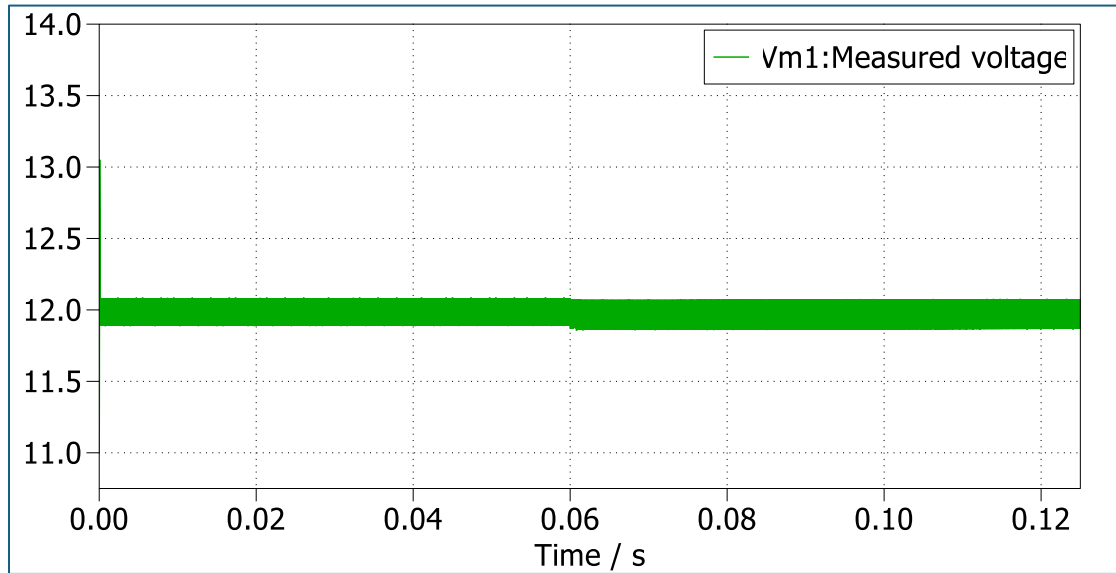


Figure 13. $K_p=11$, $K_i=16.9$, step change in voltage at $t=0.06s$

K_p	$V_{out,avg}$ before step	$V_{out,avg}$ after step	V_{out} ripple before	V_{out} ripple after	% OS	% US	Settling time	Comment
0.04	12.00 V	12.00 V	12.000-12.000=0.00V	12.01-12.00=0.00 V	0%	0%	20ms	Weak proportional action; K_i works alone, slower
11	12.00 V	11.97 V	12.08 - 11.90 = 0.18 V	12.07-11.86 = 0.21 V	0%	0%	34 μs	Clean, fast correction; tiny residual from losses

Table 8. Comparison of analytical and the optimum K_p for voltage step

Feedback Control Analysis of a Buck Converter

Key PI findings: SS error is eliminated ($T(0) = 1$ guaranteed by integrator). $z = \frac{K_i}{K_p}$ must be placed below ω_c for stable integral action — $K_p = 0.04$ gives $z = 424 < \omega_c = 1414$ causing aggression; $K_p = 11$ gives $z = 77 < \omega_c$ giving smooth correction. Increasing K_p speeds proportional response but increases post-step ripple at lower R due to reduced ζ

5.3. PID Control

Motivation: PI controller bandwidth is limited by $\omega_c = 0.1\omega_n$. Adding a derivative term (phase lead) allows a higher crossover and faster response.

Controller form: $G_c(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d(s+z_1)(s+z_2)}{s}$;

$$|G_c(j\omega_c)| \times |G_{vd}(j\omega_c)| = \frac{K_d \sqrt{(\omega_c^2 + z_1^2)(\omega_c^2 + z_2^2)}}{\omega_c} \times 24 = 1$$

Zero placement: $z_1 = 0.3 \times \omega_c = 424 \text{ rad/s}$

$$z_2 = 1 \times \omega_c = 1414 \text{ rad/s}$$

$$K_d = 1.995 \times 10^{-5};$$

$$K_p = 0.0367;$$

$$K_i = 11.97$$

$$T(s) = \frac{G_c(s)G_{vd}(s)}{1+G_c(s)G_{vd}(s)} = \frac{10^8 \left(K_p + \frac{K_i}{s} + sK_d \right)}{s^2 + 10^3 s + 2 \times 10^8 + 10^8 \left(K_p + \frac{K_i}{s} + sK_d \right)} = \frac{10^8 s K_p + 10^8 K_i + 10^8 s^2 K_d}{s^3 + s^2 (10^3 + 10^8 K_d) + s (2 \times 10^8 + 10^8 K_p) + 10^8 K_i}$$

$$s = -159.3, -3890, -92735$$

Simulation: Keep K_i and K_d fixed, and tune K_p

i. Apply a step change in load at $t=0.01\text{s}$: $10 \Omega \rightarrow 5 \Omega$

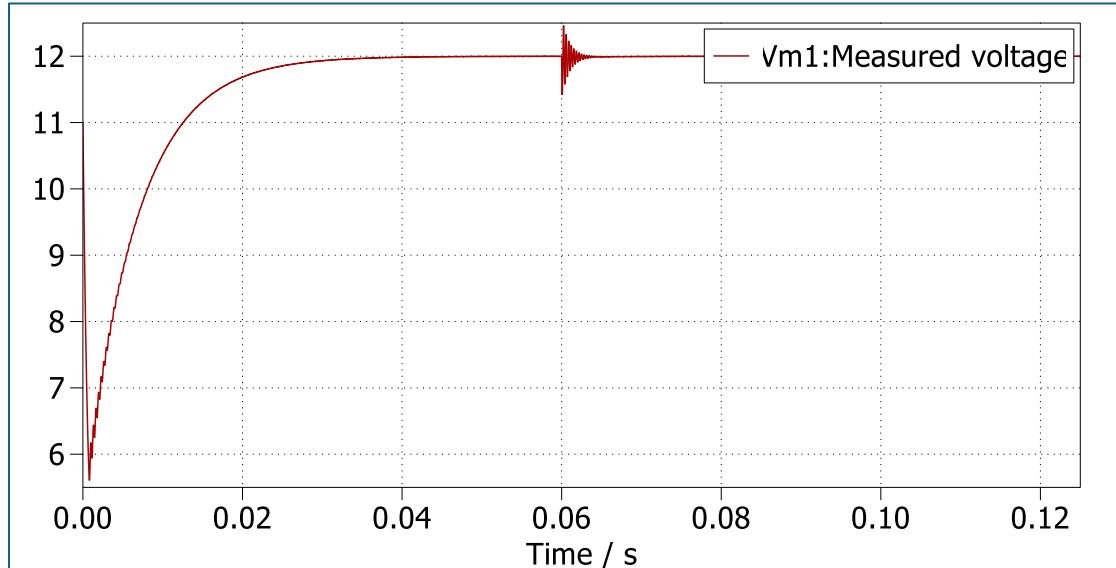


Figure 14. $K_p = 0.0367, K_i = 11.97, K_d = 1.995 \times 10^{-5}$, step change in load $10 \Omega \rightarrow 5 \Omega$ at $t=0.06\text{s}$

Feedback Control Analysis of a Buck Converter

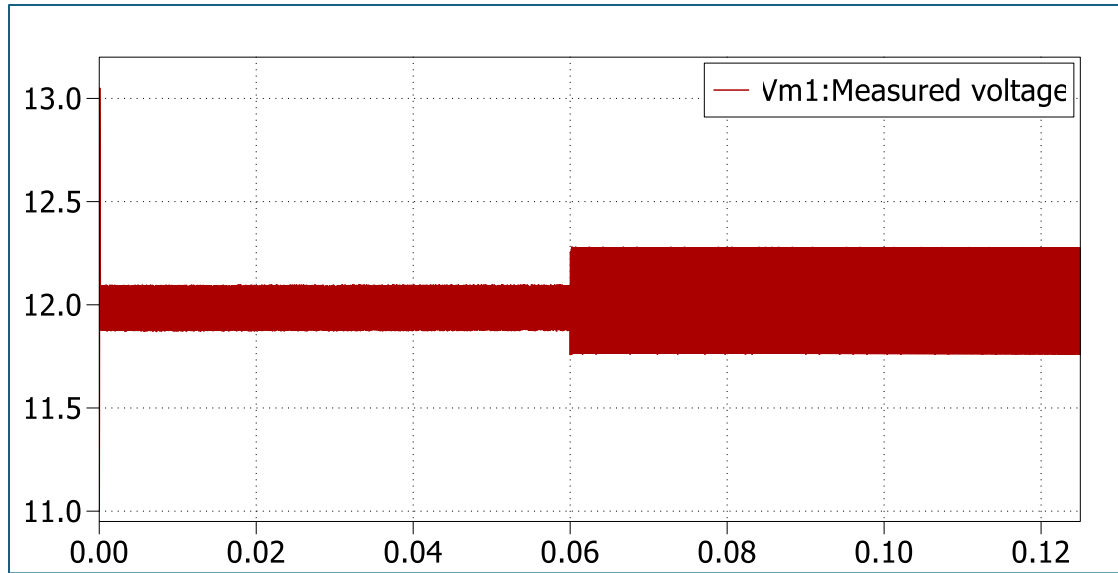


Figure 15. $K_p = 11$, $K_i = 11.97$, $K_d = 1.995 \times 10^{-5}$, step change in load $10 \Omega \rightarrow 5 \Omega$ at $t = 0.06$ s

ii. Keep the resistor at its original value, change the voltage from 24V to 20V as a step change:

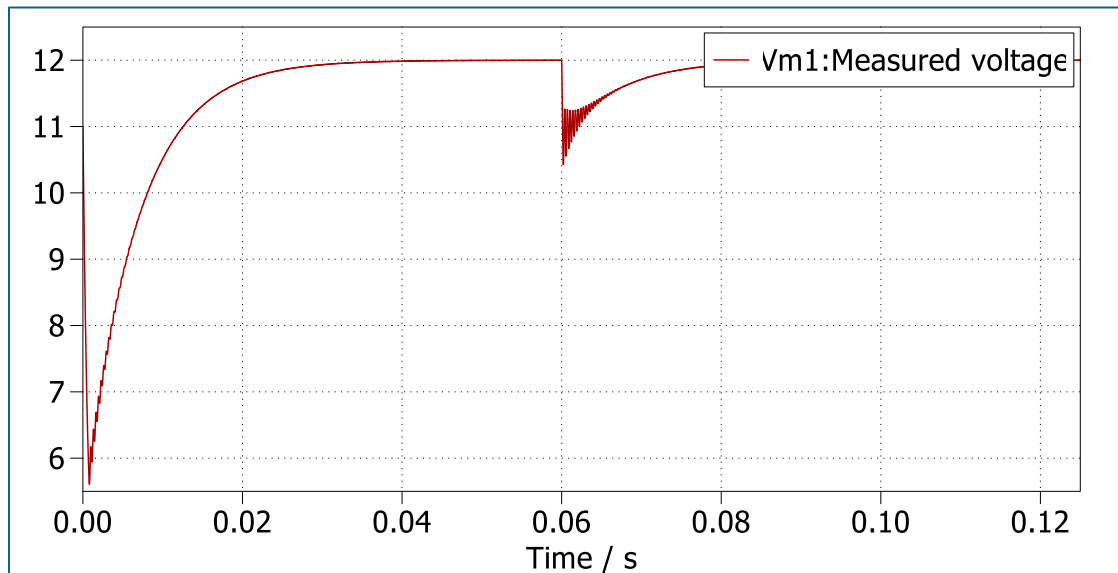


Figure 16. $K_p = 0.0367$, $K_i = 11.97$, $K_d = 1.995 \times 10^{-5}$, step change in voltage $24 \text{ V} \rightarrow 20 \text{ V}$ at $t = 0.01$ s

Feedback Control Analysis of a Buck Converter

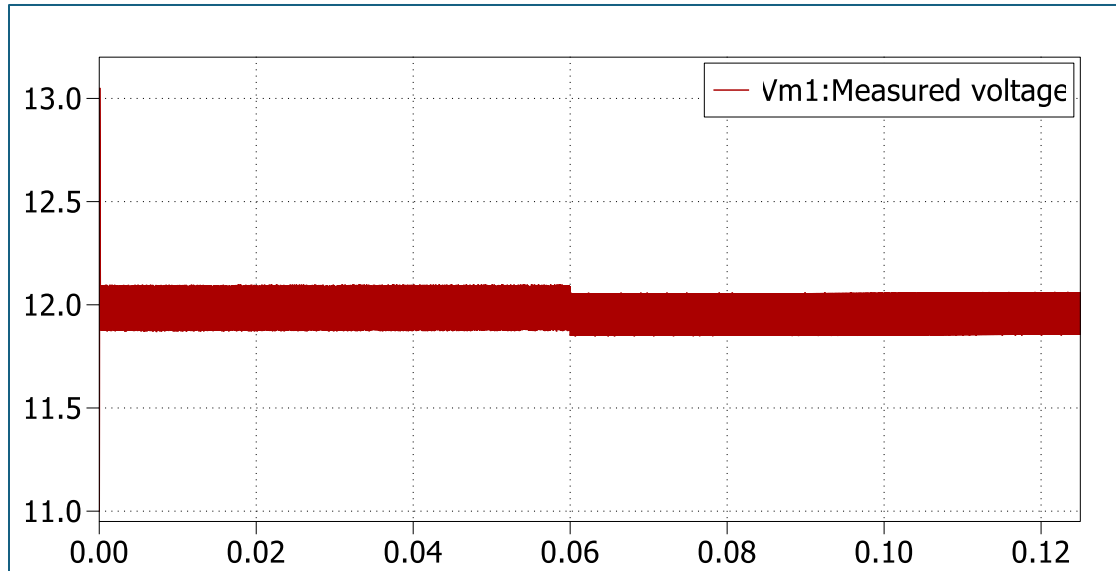


Figure 17. $K_p = 11$, $K_i = 11.97$, $K_d = 1.995 \times 10^{-5}$ step change in voltage $24\text{ V} \rightarrow 20\text{ V}$ at $t=0.06$

Test	Before (avg)	After SS	Ripple before	Ripple after	% OS	% US	Settling	Comment
load step	12.00 V	12.00 V	0.18 V	0.35 V	0%	$\approx 1\%$	< 1 ms	Fast recovery; SS error=0; derivative anticipates load change
V_{in} step	12.00 V	12.05 V	0.18 V	0.18 V	0%	$\sim 0.5\%$	< 1 ms	Rapid V_{in} correction; tight ripple; faster than PI $K_p = 11$ (9ms)

Table 9. PID results for load step and V_{in} step, $K_p = 11$, $K_i = 11.97$, $K_d = 1.995 \times 10^{-5}$

Why PID: At this design's crossover ($\omega_c = 0.1\omega_n$, identical to P and PI), PID's poles (-159.3, -3,890, -92,735) are all real, but the dominant pole (-159.3) is actually SLOWER than PI's dominant pole (-207.6). At this shared, conservative crossover, adding a derivative term provides NO speed advantage over PI — only added complexity and derivative-noise sensitivity. The genuine benefit of a derivative term only appears if it is used to push the achievable crossover frequency higher than PI can safely support — which this particular design point does not attempt.

Derivative noise sensitivity: An ideal derivative $K_d \times s$ has gain growing without bound at high frequency (+20dB/decade), amplifying switching and quantization noise. In practice a first-order rolloff filter $\frac{N}{s+N}$ should be applied to the derivative path. For this design, $N \approx 10,000\text{ rad/s}$ ($\approx 7 \times \omega_c$) leaves the closed-loop poles essentially unchanged from the ideal case, confirming the filter is safely realizable here — this is NOT true at more aggressive crossovers.

5.4. Feedforward + P — Closed-Loop Eigenvalues

Control law: $d = D + K_p(V_{ref} - V_{out})$

i. Apply a step change in load at $t=0.01\text{s}$: $10\ \Omega \rightarrow 5\ \Omega$

Feedback Control Analysis of a Buck Converter

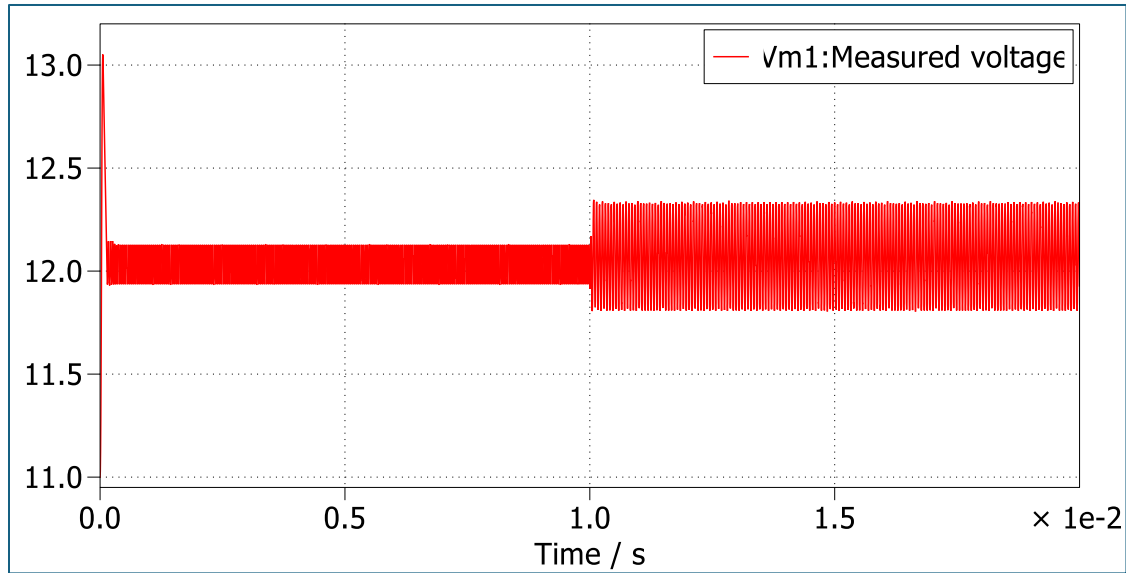


Figure 18. $D=0.5, K_p = 11$, step change in load $10\ \Omega \rightarrow 5\ \Omega$ at $t=0.01s$

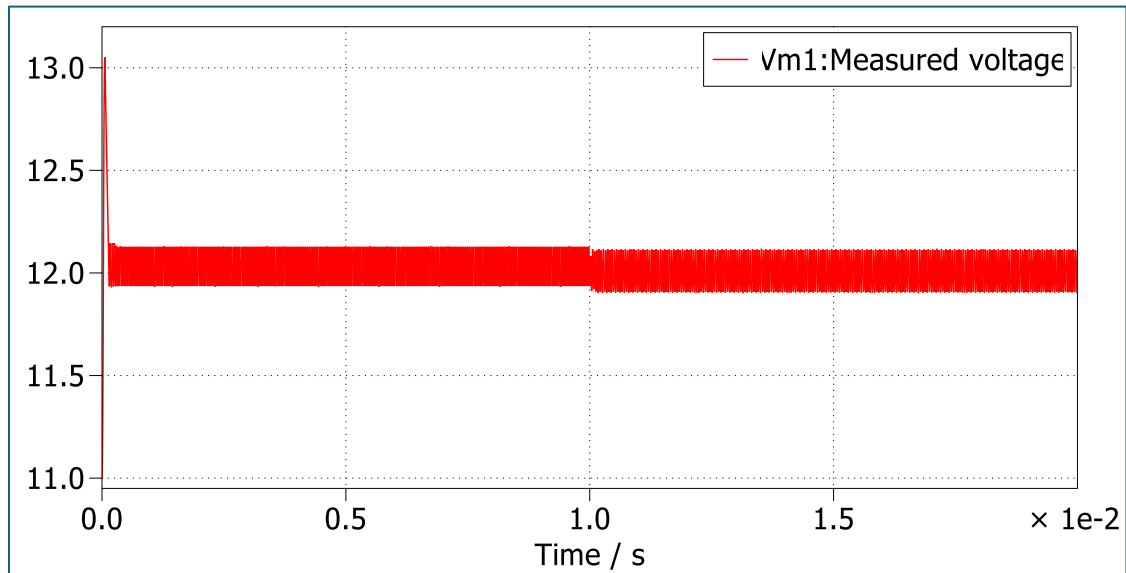


Figure 19. $D=0.5, K_p = 11$, step change in voltage $24\ V \rightarrow 20\ V$ at $t=0.01s$

Test ($K_p=11$, $D_0=0.5$)	Before (avg)	After SS	Ripple before	Ripple after	% OS	% US	Settling time (ms)	Comment
Load step $10 \rightarrow 5\ \Omega$	12.05 V	12.00 V	0.18 V	0.35 V	0%	~1%	< 1	D_0 sets nominal operating point; load step still causes ripple increase at $5\ \Omega$ since D_0 is independent of load
Vin step $24 \rightarrow 20V$	12.05 V	12.00 V	0.18 V	0.18 V	0%	~0%	< 0.5	$D_0=0.5$ fixed — does NOT compensate Vin change automatically

Table 10. Results of simulation of Feedforward + P

Benefit vs P-only: D_0 term shifts startup operating point to correct duty cycle immediately — no initial transient as controller finds the right D. Startup SS error is reduced because $d = 0.5$ when $e=0$, which is correct. However,

Feedback Control Analysis of a Buck Converter

for V_{in} changes, $D_0 = 0.5$ is only correct at $V_{in} = 24V$. Measuring V_{in} in real-time and using $D_0 = \frac{V_{ref}}{V_{in}}$ would eliminate SS error for V_{in} steps too.

Limitation: Eigenvalues unchanged from P-only — feedforward improves DC accuracy but not dynamic speed. Load step response is identical to P-only since D_0 does not change with load.

5.5. Frequency Response Analysis — Bode Plot Comparison

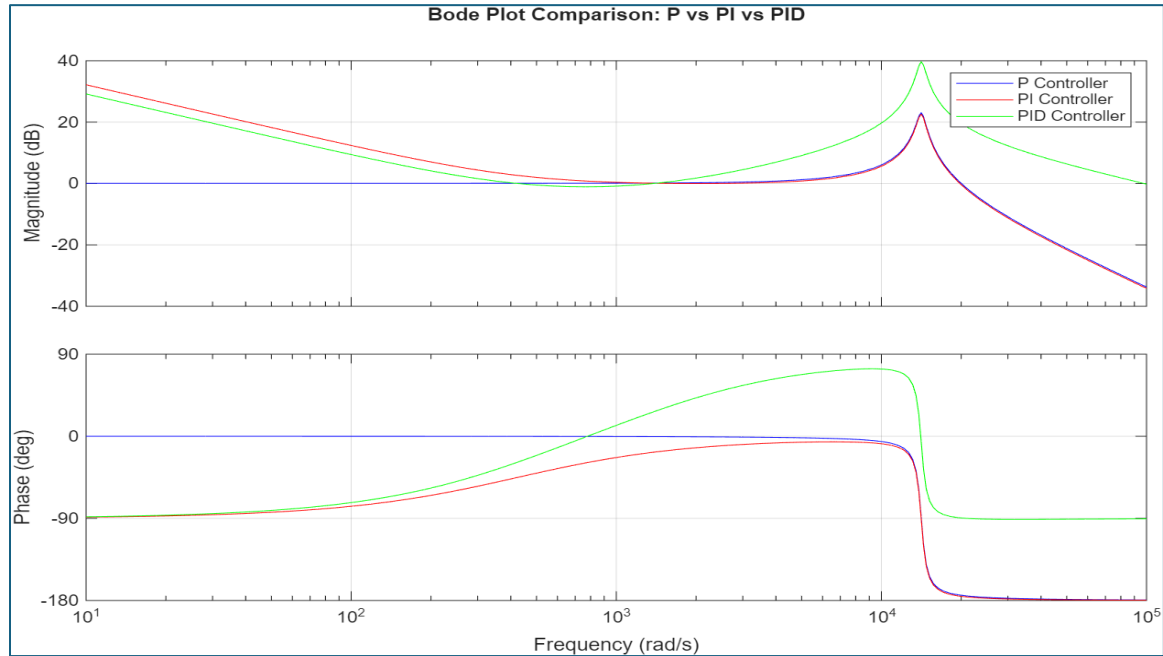


Figure 20. Bode plot for P, PI, & PID

What the Bode plot shows for each controller

P controller: flat low-frequency gain set by $K_p \cdot G_{vd}(0)$; resonance peak near $\omega_n = 14142 \text{ rad/s}$ ($\approx 23 \text{ dB}$); no phase compensation, so phase margin is entirely determined by the plant.

PI controller: +20dB/decade low-frequency slope from the integrator, forcing infinite DC gain and zero SS error; the PI zero recovers some phase before crossover but adds nothing at high frequency.

PID controller: derivative term adds phase lead peaking near $\sim 70^\circ$ and critically shifts high-frequency phase to settle at -90° (not -180° like P/PI, since the derivative's own $+90^\circ$ combines with the plant's -180°). This is the specific mechanism that rescues PID's far-frequency crossing.

Controller	0dB crossings(rad/s)	Worst-case Phase Margin	
P	20018	5.70°	Marginally stable — far below 45° safe practice
PI	1793, 2294, 19807	4.66°	Marginally stable at high-freq crossing
PID	426, 1382, 97948	89.52°	Robust at every crossing

Table 11. Stability margins (full-spectrum sweep)

6. Summary and Overall Conclusion

Feedback Control Analysis of a Buck Converter

Controller	Gains	T(0)	SS Error	Settling	Eigenvalues	PM (worst case)
Open-loop	$D_0 = 0.5$	--	0.46 V	N/A	$-500 \pm j14,142$	N/A
P design	$K_p = 0.042$	0.5	6.0 V	7.5 ms	$-500 \pm j19,994$	5.70°
P ($K_p=11$)	$K_p = 11$	0.996	0.046V	< 1ms	$-500 \pm j230,217$	0.25°
FF+P ($K_p=11$)	$K_p = 11, D_0 = 0.5$	0.996	0.05V	< 1ms	$-500 \pm j230,217$	0.25°
PI design	$K_p = 0.0399, K_i = 16.9$	1.0	0V	$490\mu s - 1.5ms$	$-207.6, -396.2 \pm j19,780$	4.66°
PI ($K_p=11$)	$K_p = 11, K_i = 16.9$	1.0	0V	$34\mu s - 1.73ms$	$-207.6, -396.2 \pm j19,780$	0.25°
PID design	$K_p = 0.0367, K_i = 11.97, K_d = 1.995 \times 10^{-5}$	1.0	0V	25ms	$-159.3, -3,890, -92,735$	89.52°
PID ($K_p = 11$)	$K_p = 11, K_i = 11.97, K_d = 1.995 \times 10^{-5}$	1.0	0V	<1ms	$-1.08, -48,380 \pm j225,076$	23.80°

Table 12. Summary

Feedforward eliminates SS error only when $D_0 = \frac{V_{ref}}{V_{in}}$ is accurate (correct V_{in} assumed).

P-control: Fast proportional response but inherent, non-zero SS error ($T(0) < 1$ always). Settling time is fixed at $\sim \frac{4}{500} = 8ms$ envelope regardless of K_p , since the real part of the closed-loop poles is set entirely by R and C. Critically, the true stability margin (5.70°) is far worse than the pole locations suggest, due to an unintended resonance-driven second crossing.

Feedforward +P: Identical dynamics to P-only (same eigenvalues) but D_0 term pre-sets duty cycle so $e=0$ gives $d = D_0$ exactly. Eliminates SS error without an integrator when V_{in} is known.

PI control: Integrator guarantees $T(0)=1$ for any K_p — zero SS error is structural, not tuned. However, PI shares the same hidden phase-margin vulnerability as P-control (4.66° at the high-frequency crossing), since it has no mechanism to compensate for the plant's resonance-driven phase loss.

PID control: At the shared design crossover $\omega_c = 0.1\omega_n$, PID offers no speed advantage over PI — its dominant pole is actually slower. Its real, demonstrated advantage is stability robustness: PID is the only controller whose derivative-driven phase lead extends far enough in frequency to rescue the unintended high-frequency crossing induced by the plant's resonance, achieving 89.52° phase margin versus $\frac{P}{PI's} \sim 5^\circ$.

Central findings of this project:

Tuned control gain $K_p = 11$ produces phase margins under 0.3° for P and PI, but the PLECS switching simulations (Tables 4-9) show well-behaved, non-oscillatory responses at this same gain. The answer lies in where these crossings occur. Averaged small-signal models are generally considered trustworthy only up to roughly $\frac{\omega_{sw}}{10}$ (here, $\approx 125,664$ rad/s) — beyond that, the assumption that PWM switching can be treated as continuous averaging breaks down. The $K_p=11$ crossover frequencies ($\approx 230,260$ rad/s for P/PI) sit well past this guideline ($\approx 18\%$ of the 200kHz switching frequency), meaning the linearized Gvd(s) model is not a reliable predictor of stability there. The near-zero phase margins are a real feature of the continuous-time model, but that model is being extrapolated outside its region of validity — the actual switched converter has additional dynamics (PWM sampling, zero-order-hold effects) that the averaged model does not capture at

Feedback Control Analysis of a Buck Converter

this frequency, which is why the switching-level PLECS results remain stable despite the alarming linear prediction.

The standard single-frequency crossover design method ($|G_c(j\omega_c)Gvd(j\omega_c)|=1$) is INSUFFICIENT on its own for lightly-damped plants ($\zeta=0.0354$ here). Because the resonance peak boosts loop gain well above 0dB near ω_n regardless of how conservatively ω_c is chosen, an underdamped plant can produce an unintended second (or third) 0dB crossing near resonance, where phase has already collapsed toward -180° . A design that appears safe from pole locations and a single design-point magnitude check can still be marginally stable in practice. This method must therefore be paired with a full-spectrum phase-margin verification (e.g., a complete Bode sweep or MATLAB's margin function checking ALL 0dB crossings, not just the intended one) before a design can be considered genuinely robust.